

Matched-Budget Comparison with Conjugate Gradients

Softmax-Kernel In-Context Bayesian Near-Field LSM

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5 August 2026

Abstract

At equal depth, prompt-conditioned PCG is substantially more accurate and better calibrated than unpreconditioned CG on the tested strongly ill-conditioned near-field systems. At context $m = 48$ and $T = 32$, it reduces the physical-coordinate posterior-mean residual by $2.05\times$, reduces the relative LSM-score error by $6.44\times$, and raises numerical 95% Bayesian coverage from 0.243 to 0.873. This is an iteration-efficiency and UQ result: localization average precision is almost unchanged because it is already limited by the physical point-spread function. The learned method is not the strict wall-clock winner against a correctly specified, training-free angular-Jacobi PCG control.

Equal operator-application budget

All iterative methods below use the same two posterior right-hand-side blocks and $T = 32$ transformed-operator applications. Timings are measured per held-out batch on the same implementation and hardware.

Method	Physical residual	CG gain	Score error	AP	95% coverage	ms
identity-CG	1.067	1.00	0.0292	0.802	0.243	10.46
population-PCG	0.673	1.58	0.00533	0.805	0.854	12.14
context-PCG	0.521	2.05	0.00454	0.805	0.873	13.75
hybrid-PCG	0.630	1.69	0.00569	0.805	0.835	13.21
angular-Jacobi-PCG	0.629	1.70	0.00550	0.805	0.851	9.84
looped HB	0.983	1.09	0.139	0.478	0.095	83.62
looped Richardson	1.684	0.63	0.293	0.304	0.092	83.71
looped Chebyshev	2.28×10^3	< 0.01	0.157	0.617	0.175	86.24

The raw median condition number is 2.66×10^4 . Context-PCG reduces it by $23.22\times$ to 1.09×10^3 ; analytic angular-Jacobi reduces it by $24.28\times$ to 1.01×10^3 . The normalized commutator between the fixed GP basis and the task Hessian is 0.221, exposing the remaining inability of modal gains to rotate obstacle-dependent eigendirections.

Accuracy, UQ, robustness and time

On independent acquisition geometries, the paired geometric-mean physical residual gain CG/context-PCG is 2.241 (95% log-gain CI 2.169–2.316), and context-PCG wins 96.4% of 768 paired batches. Its paired numerical-coverage increase is 0.728 (0.718–0.737). In contrast, the paired average-precision increase is only 0.0021 (0.0015–0.0027). Learned conditioning therefore restores numerical posterior fidelity without materially changing physical localization resolution.

The stress audit uses 16 marginal and four joint-shift levels, with 192 independent acquisition batches per level. Context-PCG wins at least 62.5% of pairs at each level, with gains from $1.07\times$ to $4.32\times$, and is best on 11/20 levels. At the extreme joint shift it retains $1.96\times$, versus 1.34 for angular-Jacobi, 0.42 for population-PCG and 0.12 for looped HB; its numerical-coverage gain nevertheless falls to zero.

For residual target 0.01 at $m = 48$, context-PCG uses $T = 96$ and 23.13 ms, whereas the shared-architecture CG uses $T = 128$ and 24.91 ms. Stripped optimized CG needs 22.86 ms, and angular-Jacobi PCG is fastest at 19.24 ms. Across all 18 context/target cells, angular-Jacobi is fastest in 11 and context-PCG in none. Hence the learned factor is valuable when conditioning is unknown and iteration budget is constrained, but it does not yet amortize its controller overhead better than the correct analytic GP-basis factor on this small dense problem.

Verdict — Versus ordinary CG: yes, clearly better as a solver and for numerical Bayesian UQ. Versus informed analytic PCG: not yet a universal speed or high-accuracy win. A stronger next architecture should learn task-dependent eigendirection rotations or a block/polynomial operator, not merely enlarge the fixed-basis modal controller.

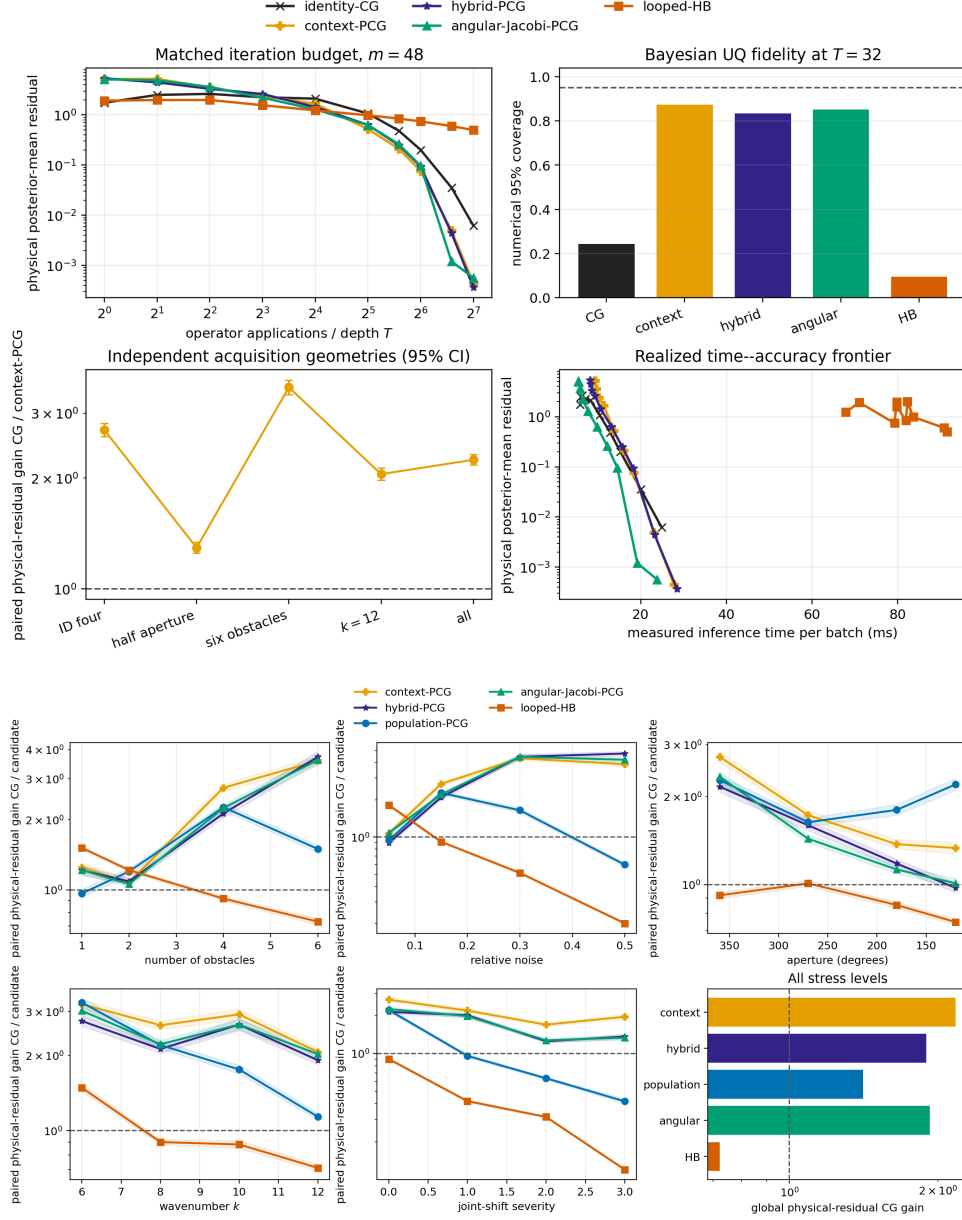


Figure 1: Top: direct physical-residual CG comparison at extrapolative context $m = 48$; the paired-geometry panel uses 768 batches. Bottom: paired physical-residual gains over continuous physical stress axes, with 95% mean-log-gain intervals. Values above one favor the preconditioned method over CG.